GATE TRIAGE FOR CONTAINER TERMINALS

Rung

Keep the gate open when the data goes bad.

Code
github.com/Hratchg/rung

Demo video
Release v1.1

The problem

Every truck leaving a container terminal goes through the gate. Five systems describe each box, and they disagree all the time.

On a bad night the safe move is to close the gate, and every truck in line pays for it.

191 boxes waiting to leave Pier 7 tonight

129 trucks booked through 6 gate lanes

How it works

1 Five sources

Carrier · Terminal · Customs · Weighbridge · Reefer

Read for every box, each field badged live or simulated.

2 Trust and conflicts

Trust fades as a reading ages and drops 30 percent for each conflict.

Weight · Records · Reefer · Dark feed · Clock

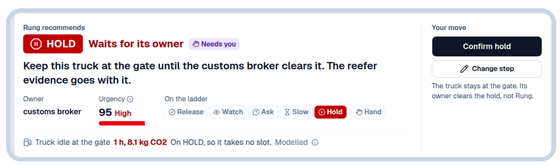
3 The five step ladder

Hold	Waits for owner
Slow	New slot
Ask	Fresher facts
Watch	Recheck
Release	Cleared

Hand A person decides

15 small rules vote. The gentlest safe step wins.

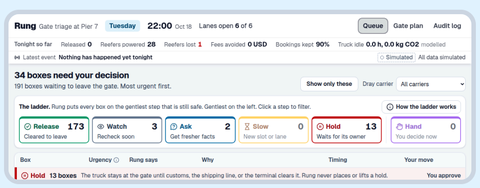
4 The planner approves



Every hold and hard release waits for one tap, and lands in the audit log with its reason.

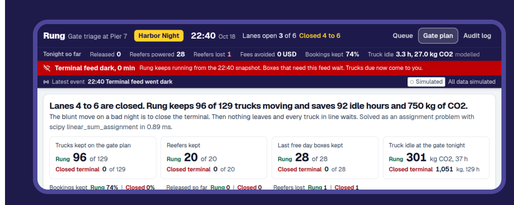
Dual use: one app, two nights

An ordinary Tuesday



- A dispatcher's pull list: 34 of 191 boxes need a decision.
- The voice agent calls truckers about weight mismatches.

Harbor Night



- Half the lanes close, a customs hold lands, a beef reefer warms, the feed goes dark.
- Rung re-plans every truck in under 1 ms.

Harbor Night results

Lanes 4 to 6 down, feed dark. Rung against closing the terminal.

96 of 129
trucks kept moving
0 if the terminal closes

20 of 20
reefers kept
0 if the terminal closes

28 of 28
last free day boxes kept
0 if the terminal closes

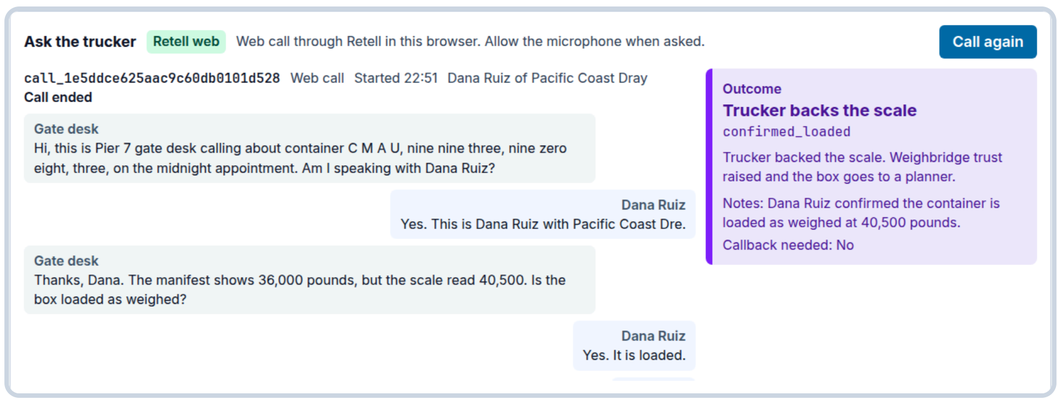
about **750** kg
truck idle CO2 avoided
92 idle hours, modelled

The Retell voice agent

Manifest 36,000 lb, scale 40,500 lb. Rung asks.

1. Pier 7 gate desk, a Retell voice agent, calls the trucker.
2. The trucker backs the scale, one of four fixed outcomes.
3. The box moves to **Hand** and the planner decides.

The model runs the call. Rung's rules decide what happens next.



A real Retell web call, with its call id in the audit log.

Go global

Global by design. The decision core never changes: one ladder, one trust model, one gate optimizer, and a person approves every hold. A region profile swaps the edges:

	Los Angeles	Rotterdam	Shanghai
Data	Terminal49	Portbase	China Single Window
Units	lb, USD	kg, EUR	kg, CNY
Customs	CBP	Douane	China Customs

Lagos, where trucks queue for days, needs it most.

Federal funding pathway

Launch PRIMARY — **NSF SBIR Phase I**, America's Seed Fund (NSF 26-510). Up to $305K over 6 to 18 months to prove Rung on live San Pedro Bay terminal data.

Harden ALTERNATIVE — **DoD SBIR** with USTRANSCOM and SDDC for strategic seaports such as Long Beach.

Scale ALTERNATIVE — Port-led **MARAD PIDP** grants.

One tool for ordinary Tuesdays and Harbor Nights.

Team: Hratch Ghanime

Pier 7 is simulated with a fixed seed. The voice call is real.
CO2 is modelled at 8.14 kg per truck idle hour (DOE and Argonne, EPA).
Built with FastAPI, scipy, React, Retell AI, and Terminal49.